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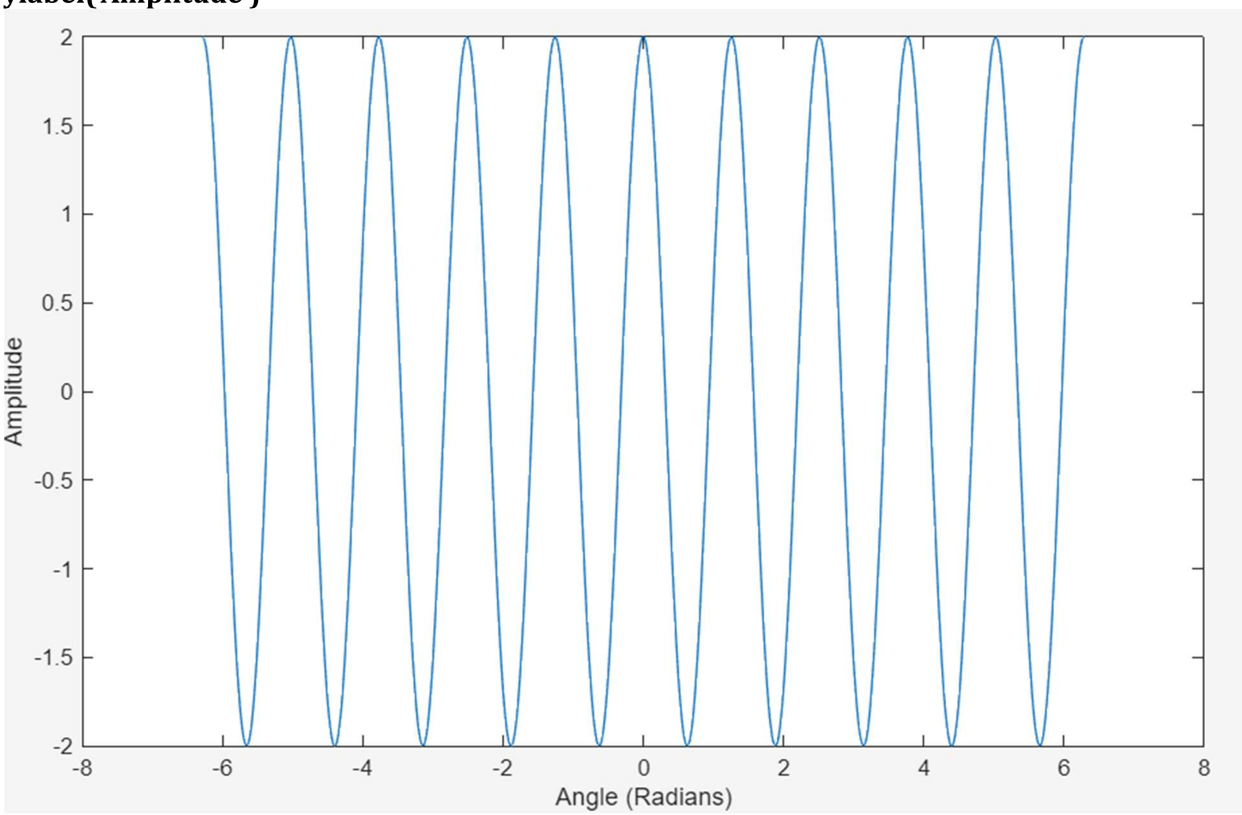
EEL 3135 – Classwork #01

Question #1: Consider the signal x(t) = $2cos(5t)$

(a) Graph $x(t)$ for $-2\pi \leq t \leq 2\pi$ using MATLAB. Provide code and results.

```
figure(1)
t = -2*pi:pi/200:2*pi;
y = 2*cos(5*t);
plot(theta, y)
xlabel('Angle (Radians)')
ylabel('Amplitude')
```



Graph a shifted version $x(t-1)$ for the same interval using MATLAB. Provide code and results.

(b) Explain how the shift affects the phase of the signal.

It shifts the wave to the right by 1.

(c) Compare the frequency and amplitude of the two signals.

The frequency and the amplitude are the same.

Question #2: Phasors and Complex Representation

A sinusoid is given by $y(t) = 3\cos(10\pi t + \pi/6)$.

(a) What is the angular frequency, conventional frequency, and phase of y(t)?

10pi, 5Hz, pi/6 phase.

(b) Write the phasor representation

$$Y = Ae^{j\phi}$$

$$\boldsymbol{Y = 3e^{j(\pi/6)}}.$$

(c) Determine the real and imaginary parts of

$$Y = e^{j\omega_0 t}.$$

Real: 2.598, Imaginary: 1.5

(d) Two phasors are defined as

$$z_1 = 2 + j, \qquad z_2 = -1 + 3j.$$

Find their sum

$$z_s = z_1 + z_2 = 1 + 4j$$

and express in both rectangular and polar form.

Rectangular: **1 + 4j, Polar: sqrt(17)phasor(75.96)**

Question #3: Simplification Using Trigonometric Identities

(a) Consider the sinusoid

$$x(t) = 8\cos\left(6\pi t - \frac{\pi}{3}\right).$$

Determine the amplitude A, angular frequency ω, and phase ϕ such that

$$x(t) = A\sin(\omega t + \phi).$$

(b) Simplify the following sum into the form $Acos(2\pi ft + \Phi)$:

$$4cos(20\pi t + \pi/3) + 2cos(20\pi t - \pi/3)$$

(c) Simplify the product $cos(12\pi t)cos(6\pi t)$